Markets without a platform

the loop economy on Swarm

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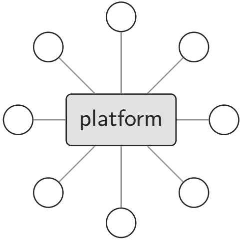
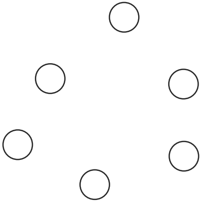
loopmarket · ontodag · recordstore · factbond github.com/petfold

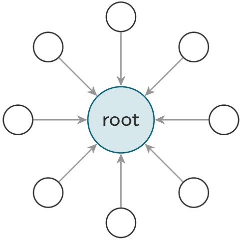
Two failure modes, one cause



the platform is the market
rent, gate, ranking



every shop an island
no discovery, no liquidity



the market is a root
anyone recomputes it

- Either the market is somebody's property, or there is no market at all.
- Missing piece: a *shared, unowned* market structure.

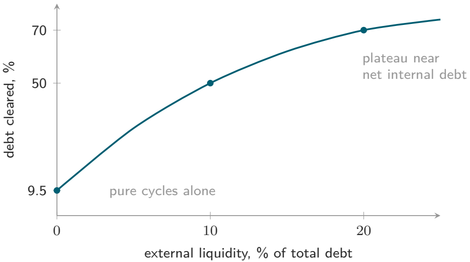
The rent, the gate, the ranking

- Take rates: app stores $\sim$30%; marketplaces 15%+ referral, plus logistics and ads. [verify]
- Gatekeeping: delisting is unilateral and unappealable.
- Ranking: the platform's algorithm decides who exists.

Platform power is structural: whoever holds the only index can charge for inclusion and shape the market by omission.

The deeper limit: everything must pass through money

- Trade needs a double coincidence of wants, or a common medium.
- Where the medium is thin, good trades don't happen.
- Time banks stall in every town: within one service type the web of offers is too sparse to close cycles.



1.28M Italian B2B invoices (arXiv:2507.22309, Fig. 10): a small bridge of liquidity turns a cycle-only regime into near-full clearing. Points are read from the paper's figure; the curve is a guide.

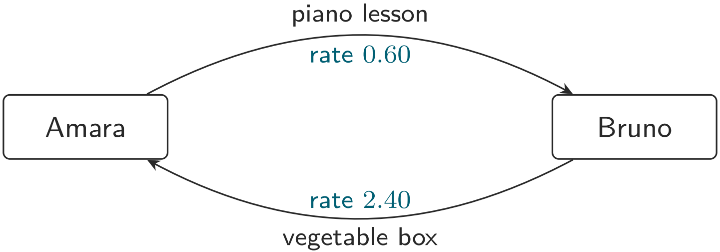
One uniform offer

offer	id = sha256(canonical bytes)
maker	amara (= feed owner)
side	give
thing	{piano-lesson} unit: course
price	100 on Amara's own scale
service	2026-09-01 .. 2027-01-01
region	disc (46.05, 14.50), r = 5 km
valid	2026-09-01 .. 2026-10-01
pins	catalogue root 05ca09... registry 4.1
carried	bond, oracle, arbitrator (enforced in P3)

Every intention has this one form. Exactly one side is a thing, the other is the maker's own scale; a want is the same record with the side flipped.

- A thing, a time, a place, a validity, a price.
- Give or want: the same form.
- Priced on the maker's *own scale*: nothing is held or transferred; the numbers cancel inside a loop.
- A shop, a person, a bus with empty seats, a stablecoin: all makers.

The simplest loop



$0.60 \times 2.40 = 1.44 > 1$ the loop is profitable

The simplest loop, on two scales

$60/100 = 0.60$

Amara

gives piano lesson	100
wants vegetable box	120

on Amara's scale

Bruno

gives vegetable box	50
wants piano lesson	60

on Bruno's scale

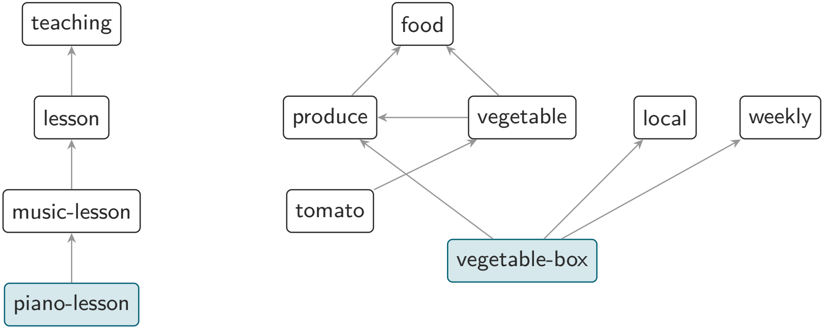
$120/50 = 2.40$

Each rate is the taker's bid over the giver's ask,
in mixed units. Around the loop the units cancel:

$$\prod \text{rate} = \prod_i \frac{\text{bid}_i}{\text{ask}_i} = \frac{120}{100} \cdot \frac{60}{50} = 1.44$$

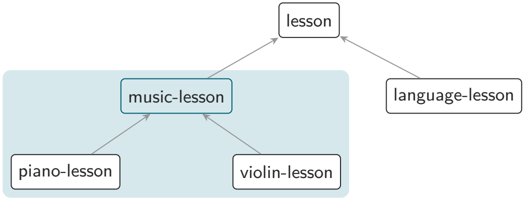
Nothing is held or transferred on either scale.

The catalogue: meanings, minutes and map cells, one order



one relation, $\sqsubseteq$ (fits within). A want for *music-lesson* is met by a give of *piano-lesson*; a want for {produce, local, weekly} is met by *vegetable-box*. Multiple parents are normal: a DAG, not a tree.

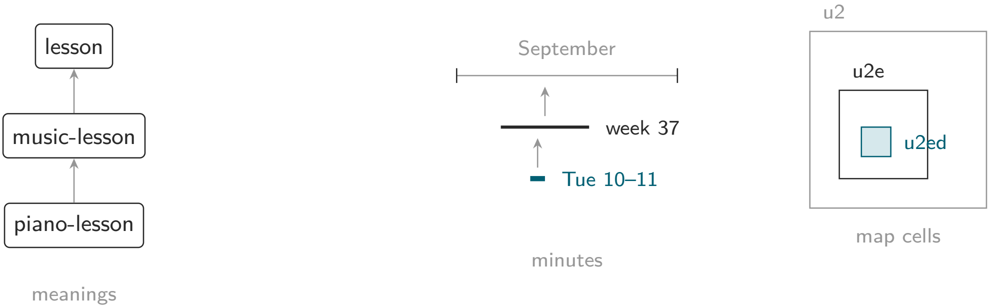
ontodag as the language of offers



Chen wants *music-lesson*: anything in its cone satisfies her.
Amara gives *piano-lesson*: inside the cone ⇒ match.
Unknown category ⇒ never matches (fails closed).

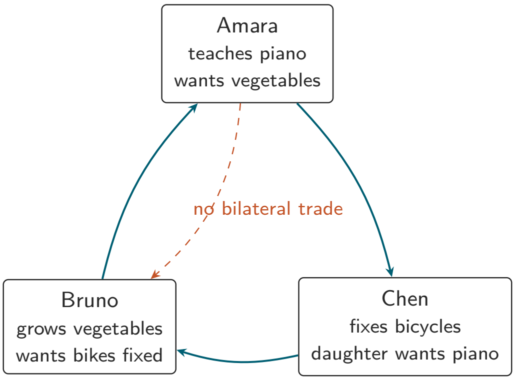
- One primitive: intersection of descendant cones.
- Shipped this week: the `core` pack, 4,137 consensus categories (WordNet, SUMO, OpenCyc, Wikidata, schema.org as witnesses) + ten domain packs.
- A catalogue is a *pinned root*: every offer names the version it was written against.
- Honest footnote: goods are rich (toaster, jeans, tomato); services are the next pack.

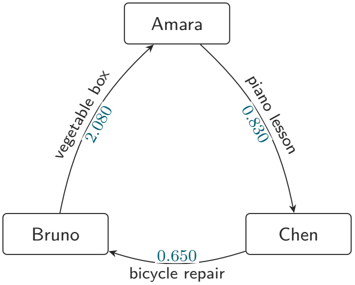
Fits-within, three ways



the same relation everywhere: a piano lesson is a lesson, Tuesday 10–11 is in week 37, cell u2ed is in u2e. The catalogue orders all three by fits-within; exact time and geo checks remain the truth at match time.

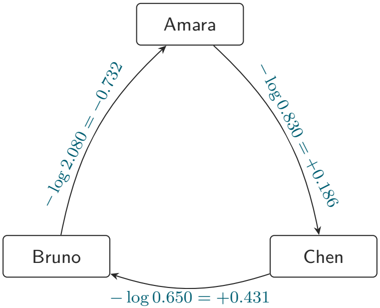
Loops: the trade nobody asked for





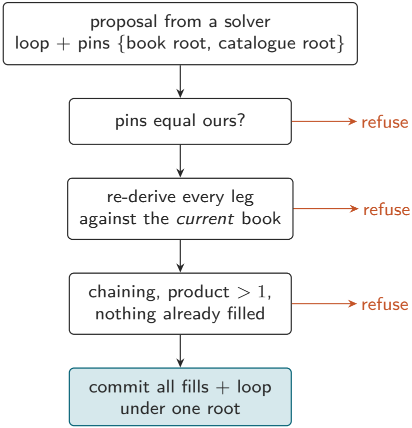
no pair can trade $0.830 \times 0.650 \times 2.080 = 1.122$ surplus 12.2%

A profitable loop is a negative cycle



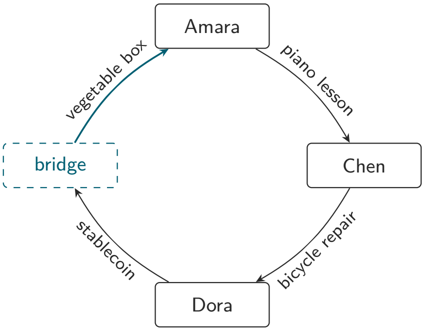
weights $w = -\log(\text{rate})$; $\prod \text{rate} > 1 \iff \sum w < 0$
a profitable loop is a negative cycle; Bellman–
Ford finds it in sorted order, deterministically

Settlement trusts no one



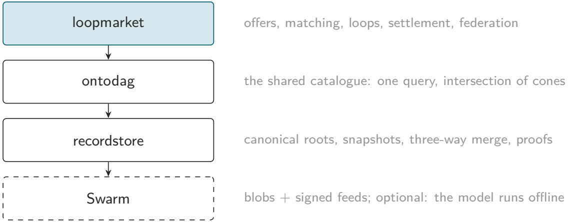
the solver is a hint, never an authority. Atomicity is one recordstore commit: either every leg fills or none does.

Money is just another offer



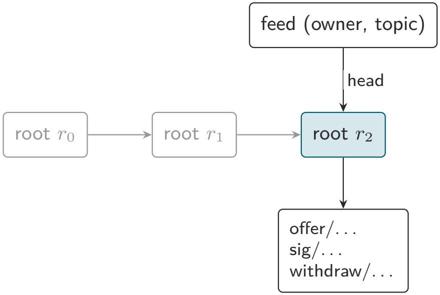
Dora can only pay in money; Amara wants vegetables. A *bridge* is an ordinary maker whose thing is a currency: it buys the vertical's staples for stablecoin and sells them at a spread. Almost-loops failing on a money leg become loops.

The stack, and why it is Swarm-shaped



- Book = versioned keyspace, canonical roots: equal content $\Rightarrow$ equal reference.
- One book per maker, under the maker's own feed and signer. Identity *is* the feed owner.
- Everything so far runs on a *light* node.
- Postage TTL is the offer's lifetime; withdrawal is a tombstone, never a delete.

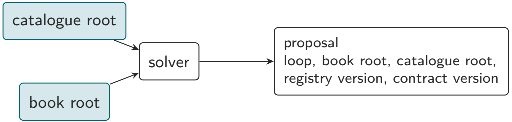
A maker's book is a feed



one maker, one feed, one signer. Identity is the feed owner address. Every version is a content-addressed root; a withdrawal is an added tombstone, never a delete; the postage stamp's TTL is the offer's real lifetime.

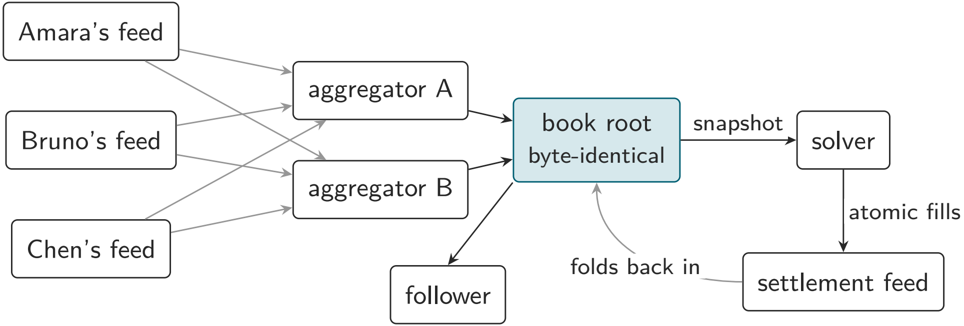
Invariants the tests refuse to let you break

- U1 uniform offer form
- U2 immutable, content-addressed
- U3 settlement trusts no solver
- U4 solve against pinned roots
- U5 positive rates only
- U6 deterministic baseline solver
- U7 vocabulary fails closed
- U11 no partially-filled loop survives a merge



frozen snapshots in, pins out. The same book yields the same loop on every replica, and a verifier a year later can replay the decision. Reproducibility beats freshness.

The federated book



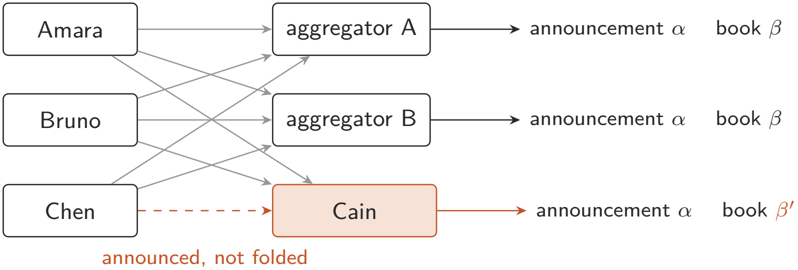
the fold is a pure function of the announced feeds: two aggregators, any order, one root. A reader may use a manifest as a cache or fold the feeds itself.

Demo: three feeds, two honest aggregators, one censor

- Three makers publish under their own feeds; two aggregators fold in different orders: byte-identical manifests.
- Mallory forges an offer in Amara's name: refused at the fold, with an attributed reason.
- Bruno withdraws: a tombstone that survives merges.
- Cain announces Chen's book and silently drops it.
- Settlement bases its own book on the fold; the loop settles; a follower reads six atomic fills from the manifest alone.

`examples/demo_federation.py` – in memory $\sim$5 s; live on a Bee light node 2–7 min (recording).

Omission is a proof, not a suspicion



same inputs claimed, different fold: evidence by construction.
audit(Cain's manifest) = Chen's two offers, each with an absence proof against β'
that anyone verifies with no store.
a solver folding the announced feeds itself lands on β.

Are aggregators necessary?

- Not for correctness, trust or permission: the fold is pure; any reader recomputes it.
- For read cost: a feed lookup costs seconds; a solver folding 1,000 maker feeds pays 1,000 lookups per beat.
- Direction: solver-side folding is the default; manifests are disposable caches.
- Discovery via a public log (Gnosis registry events as the floor, GSOC as the fast path), never via a party who can omit.

Question for this room: GSOC reception from light nodes, and pub/sub. Content-routed GSOC (one address per catalogue cone $\times$ geo cell $\times$ day) would make the index *the address space*. Stamps are not the problem; delivery guarantees are.

Roadmap

P0	P1	P2	P3	P4
in-memory prototype done	Swarm book, federation live; read path open	verifiable settle-ment on Gnosis; auctions; pricing	guarantee fabric bonds, oracles (factbond)	privacy commitments, ZK fits-within

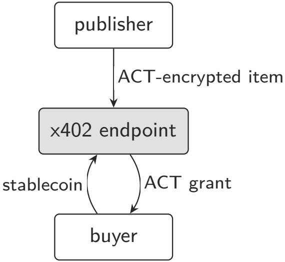
The one point of trust today is settlement. P2 removes it: inclusion and absence under the pinned book root, verified on Gnosis with recordstore's canonical-trie proofs.

Swarm AI Data Exchange and loopmarket: same substrate, same idioms

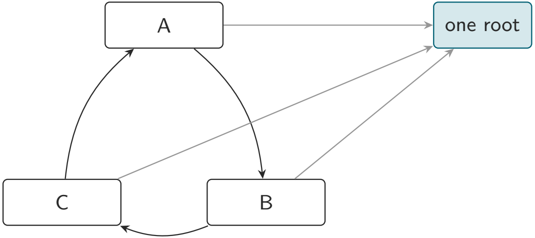
- Both: content-addressed catalogues on Swarm; an atomic root in a feed; feed owner as identity; a light node suffices to read.
- Exchange: Agent Card (ERC-8004) → catalog feed → Mantaray root → `item.jsonld`; ACT-encrypted content; x402 purchase endpoint.
- loopmarket: per-maker feed → recordstore root → offers; aggregator manifests; settlement feed.

What the exchange got right and shipped: ACT per purchase, schema.org-first vocabulary, server minimization as a principle, a working purchase flow.

A storefront and a clearing house



storefront: one seller, one buyer,
one item, one currency, one server



clearing house: k parties, no common currency,
no maker server, one atomic commit

The differences

	Swarm AI Data Exchange	loopmarket
trade shape	bilateral: one publisher, one buyer, one item	multilateral loops; no pair needs to want each other
price	set by the publisher, in a stablecoin	discovered; each maker on their own scale; surplus when $\prod$ rates > 1
medium	required (stablecoin via x402)	none required; money is a bridge offer
described	data assets (schema.org + `swarm-cat:`)	anything: catalogue conjunction $\times$ time $\times$ place
required server	the publisher's x402 endpoint	none for makers; settlement only (on chain in P2)
trust	ERC-8004 reputation + facilitator	re-verification; bonds where verification fails; reputation derived, never a gate
privacy	ACT per purchase	none yet (P4)
maturity	working purchase flow	prototype; live on Swarm; mock settlement

Why not reputation-based

- eBay: 0.3% of transactions rated negative, yet $P(\text{negative} \mid \text{partner rated negative}) > 37\%$: retaliation suppressed truthful feedback until one-sided ratings in 2007. Resnick & Zeckhauser
- Cheap ratings inflate toward uselessness. Filippas, Horton, Golden, EC'18
- loopmarket's rule U12: standing counts settled, fee-paid loops only; loss experience comes from bonded, adjudicated events (factbond).

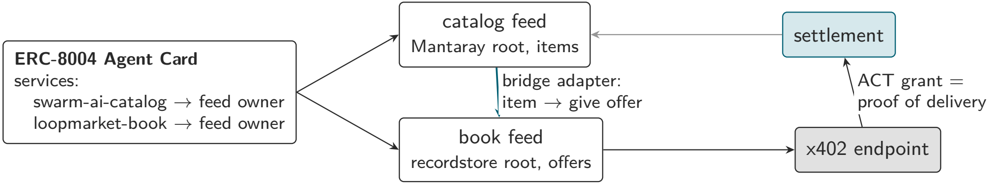
Reputation answers "should I trust this seller". loopmarket makes the question unnecessary where it can, and expensive to game where it cannot. Reputation becomes a statistic you derive, not an institution you believe.

x402 and ERC-8004 fit: as legs, identities and proofs

- **x402** is a bilateral rail; it cannot settle a k-leg loop. It fits three places the plan already has: the stablecoin leg of a bridge offer (settlement cargo, never the scoring numeraire); paid aggregator *serving*, never inclusion; solver fee collection.
- **ERC-8004 identity**: an Agent Card service entry naming the maker's book feed owner is exactly the announcement channel, with the chain registry as the censorship-resistant floor.
- **ERC-8004 reputation**: publish one-sided, aggregated signals derived from settled loops; consume feedback as one input to risk premia; never gate a trade on it.
- **ERC-8004 validation**: settlement receipts and factbond certificates are validation events with proofs.

x402 for the money leg, loopmarket for the parts x402 cannot see.

How we could converge



one identity, two services; the exchange's catalog becomes book liquidity unchanged; x402 carries the money leg and its ACT grant is the cheapest fulfilment oracle. Kept out of the shared data model: a single currency, a discovery indexer that can omit.

How we could converge

1. One ERC-8004 Agent Card carrying both a `swarm-ai-catalog` and a `loopmarket-book` service.
2. Bridge adapter: every catalog item is a give offer priced in a stablecoin; publish the catalog into a book unchanged.
3. The x402 endpoint as a fulfilment oracle: an ACT grant is a verifiable proof of delivery.
4. Shared vocabulary: the exchange's schema.org/`swarm-cat:` terms as an ontodag pack.
5. Discovery as a public log, not an indexer: keep "no synthesized server views", add "no party who can omit".
6. Keep the one-currency assumption out of the data model: a price is a number on *some* scale.

In return: ACT as a P4 Tier-1 candidate, x402 pragmatism, ERC-8004 identity, and the habit of shipping.

Asks to the room

- GSOC reception from light nodes; pub/sub timing.
- A services layer for ontodag's core pack: the goods are in; repair, tutoring, cleaning are not.
- A first vertical: digital services, agents first.

The root is the market. Swarm holds the books. The chain is the floor for discovery. The one remaining point of trust is settlement, and it is next. github.com/petfold:

loopmarket, ontodag, recordstore, factbond

Backup: a match is an edge

give Amara
{piano-lesson}, 100 on Amara's scale

$\xrightarrow[\text{rate} = 83/100]{\text{check\_match}}$

want Chen
{music-lesson}, 83 on Chen's scale

exact pairwise check: thing fits within (catalogue), windows overlap, discs intersect, both valid now, same pins
$\Rightarrow$ one edge Amara $\rightarrow$ Chen in the exchange graph

Backup: Graphviz renderings (dot2tex)